

# Yichen Benjamin Lin Zhou

## Hardware & Mechatronics Project Portfolio

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I am a mechatronics and robotics engineer completing an MSc in Robotics at the University of Sheffield, with a background in mechanical engineering (BASc, University of British Columbia). My work spans humanoid robotics R&D, autonomous competition robots, biomedical soft-robotics research, and industrial automation — from mechanical design and prototyping through to electrical integration, instrumentation, and embedded control.

This portfolio highlights a selection of my hands-on hardware projects. Each entry reflects work I personally led or contributed to across design, build, test, and integration.

### Focus Areas

#### Mechanical Design

CAD (Onshape, SolidWorks), 3D printing, prototyping, DFM

#### Electrical & Integration

Circuit assembly & debugging, wiring, sensor integration

#### Instrumentation

DAQ, force sensing, test infrastructure, statistical analysis

#### Embedded Control

Arduino/C++, PWM motor control, state machines, encoders



UBC Thunderbots RoboCup SSL robot — 2nd place, RoboCup 2024

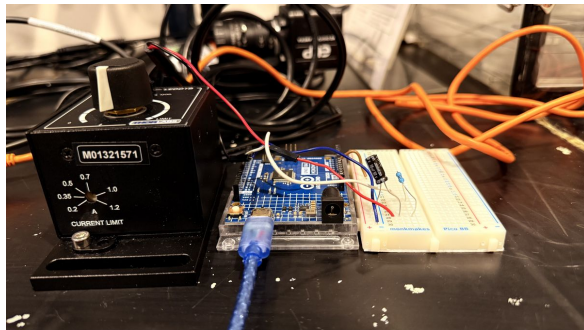
# Wireless LED Powering & Control Platform for Soft-Valve Actuation

MSc Dissertation (ELEDD02) | University of Sheffield | Supervisor: Dr. Dana Damian | 2026

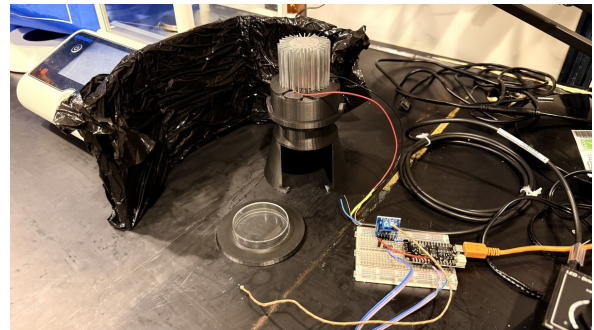
Developing hardware and control systems to optically drive light-responsive PNIPAM/graphene oxide (PNIPAM/GO) soft valves for drug-delivery and soft-robotics applications. The project combines a wired NIR LED characterisation rig — used to generate a full experimental dataset on valve actuation behaviour — with a wireless inductive power transfer (WPT) platform under development to characterise whether optical power can be delivered to NIR LEDs wirelessly, at benchtop scale, with sufficient stability and controllability.

## Wired Rig & Valve Characterisation

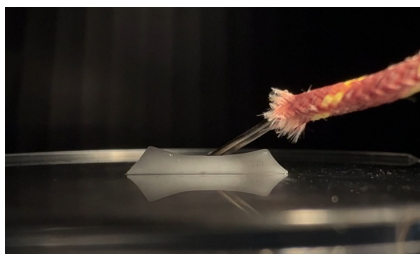
- Designed and built a NIR (810 nm) LED characterisation rig with a custom 3D-printed SM1-compatible lens adapter, Thorlabs collimating optics, and a derived irradiance model.
- Built a modular Python/Tkinter control & logging GUI with live temperature, irradiance, and camera feeds, and synchronised CSV/video export.
- Ran a full round of valve actuation testing across three PNIPAM/GO formulations (G5/G6/G7), multiple GO concentrations, and sample thicknesses — then cleaned, statistically analysed (Kruskal-Wallis testing, correlation analysis, effect sizes), and plotted the resulting dataset.



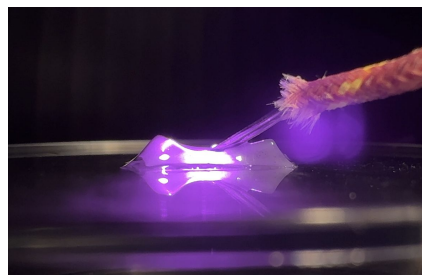
Wired NIR characterisation rig



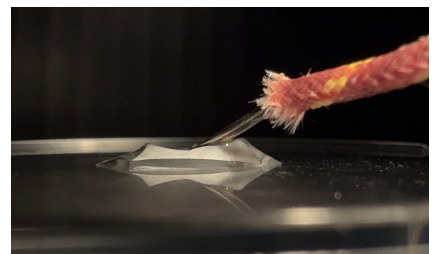
LED, lens, and custom 3D-printed adapter assembly



Before NIR exposure — valve closed



~15 s into NIR exposure — actuation in progress

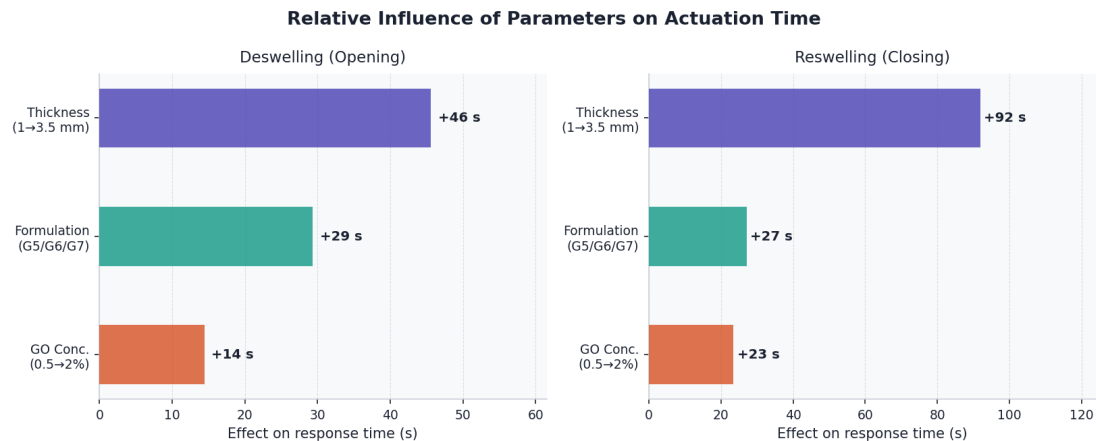


After NIR source switched off — valve open

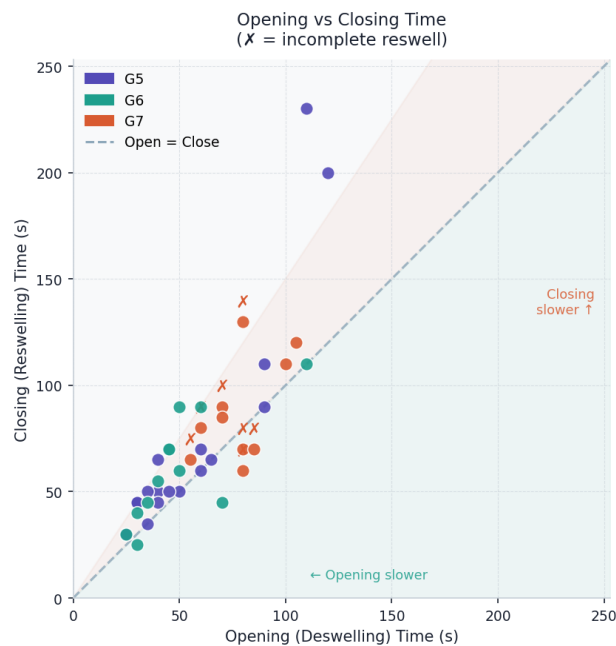
# Valve Characterisation — Key Results

MSc Dissertation (ELEDD02) | University of Sheffield | 2026

Sample thickness was the strongest and most consistent predictor of actuation speed ( $r = 0.52$  opening,  $r = 0.66$  closing), outweighing both formulation and GO concentration. Formulation had a real secondary effect — G6 fastest, G7 slowest and prone to incomplete reswelling — while GO concentration (0.5–2.0%) showed no consistent trend.



Relative influence of thickness, formulation, and GO% on response time

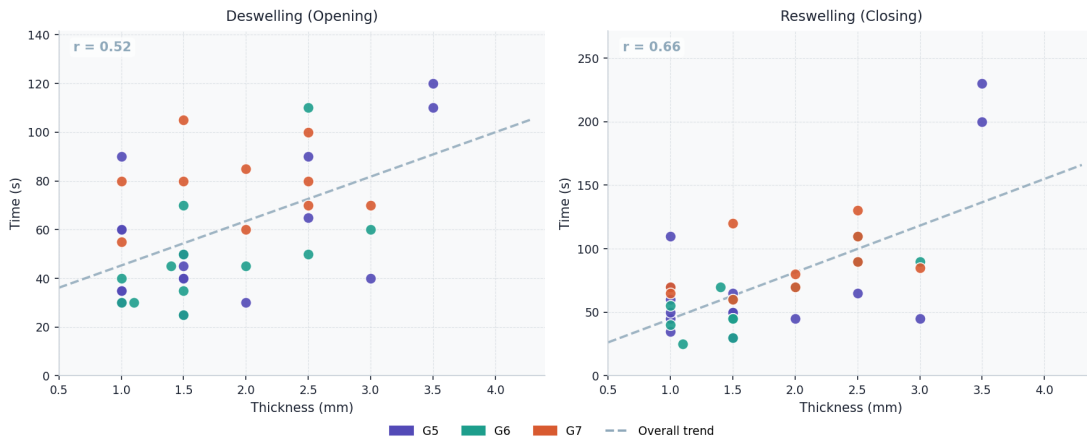


Opening vs. closing time — X marks incomplete reswell (mostly G7)

# Valve Characterisation — Key Results (cont.)

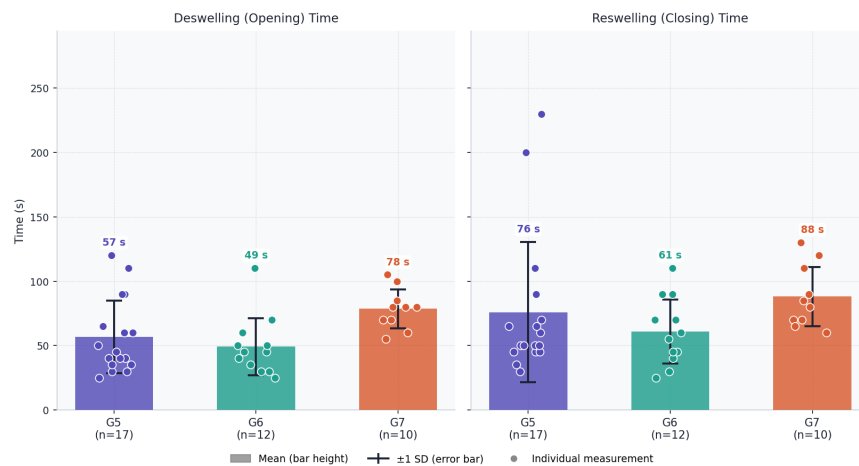
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Effect of Sample Thickness on Actuation Response Time



Thickness vs. response time by formulation

Actuation Response Times by Sample Formulation



Statistical comparison across formulation and GO concentration

## Wireless Power Platform (In Progress)

- Debugged and validated the 6-channel PSRG transmitter (30–90 kHz, AD9833 DDS via SPI); confirmed stable single-channel output.
- Fully specified a 3-coil receiver design (30/54/90 kHz) with rectification and tuning components sourced.
- Ran a power-budget feasibility analysis: identified a  $\sim 20$ – $200\times$  gap between prior wireless power delivery and the LED's full-drive requirement, and used a Biot–Savart calculation to show coil-distance reduction gives diminishing returns in this near-field-saturated regime.
- Next: wind and validate the first receiver coil, then measure delivered power across all three frequencies against the wired-rig baseline.

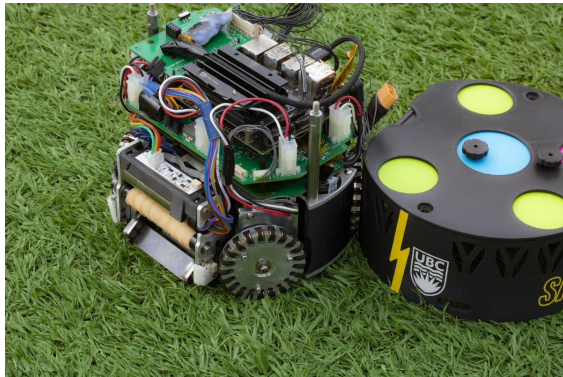


# UBC Thunderbots — RoboCup Small Size League

University of British Columbia | Sept 2019 – May 2025 | Mechanical Member → Mechanical Lead → Captain → Integration Lead

For five years I designed, built, and integrated autonomous soccer-playing robots for the RoboCup Small Size League — one of the most demanding autonomous robotics competitions in the world, where teams of six robots play fully autonomous soccer at high speed. I progressed from a mechanical team member to Captain and Integration Lead, leading a 70-member interdisciplinary team to a 2nd place world ranking (Division B) at RoboCup 2024 in Eindhoven, Netherlands.

## Competition Robots



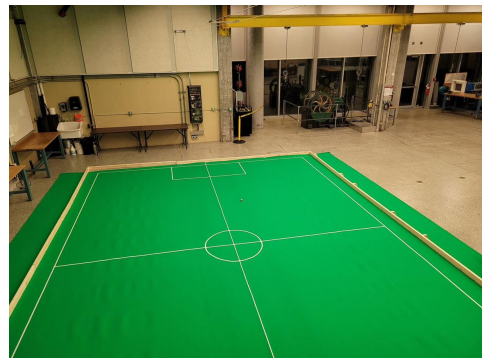
Robot with top plate removed — drivetrain, dribbler, and electronics



Robot fleet staged for competition



Autonomous match play at RoboCup 2024, Eindhoven



17 × 20 m practice field built for full-scale gameplay testing

## Role Highlights

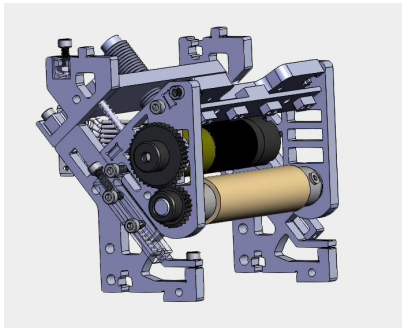
- Founded Thunderbots' first dedicated Integration Team, formalising cross-subsystem debugging and pre-competition validation of mechanical, electrical, and software systems.
- Designed test plans to verify failure points in the kicking and chipping mechanisms and to calibrate solenoid output forces.
- Led robot subsystem redesigns (dribbler & damper, drivetrain) across multiple prototype iterations — detailed in the following pages.
- Co-authored annual RoboCup Team Description Papers (2019–2024) as part of the league qualification process.

# Dribbler & Damper Redesign

UBC Thunderbots | Jan 2020 – Jul 2022 | Mechanical Lead

I led the redesign of Thunderbots' mechanical ball-control system — the dribbler and damper assembly responsible for capturing and controlling the ball at speed. The design I led used an aluminium frame rotating about a pivot point with a threaded polyurethane roller bar, optimised for reliable ball contact across a range of approach angles. This became the first prototype of the current generation of the team's dribbler.

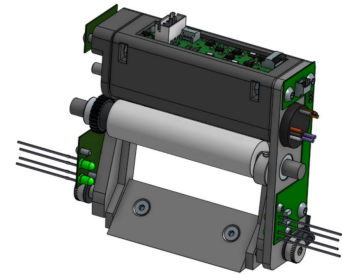
## Design Evolution



*Previous-generation dribbler (baseline)*



*Redesigned dribbler concept (CAD)*



*Refined production CAD*



*CNC-machined aluminium prototype assembly*

## Contributions

- Led a team of 5 through the full design–build–test cycle.
- Designed four prototype iterations, culminating in the competition design (Jul 2022).
- Manufactured and assembled prototypes using SolidWorks, 3D printing, and machine tools.
- Generated and reviewed engineering drawings for each iteration.

# Drivetrain & Omni-Wheel Redesign

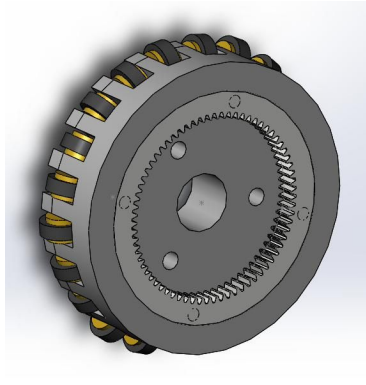
UBC Thunderbots | Sept 2019 – Jul 2022 | Mechanical Member → Mechanical Lead

I redesigned the robot drivetrain to improve overall manoeuvrability and structural rigidity for Division B competition. The work centred on re-engineering the omni-wheel geometry for greater agility and reduced mechanical play, alongside supporting chassis rigidity improvements.

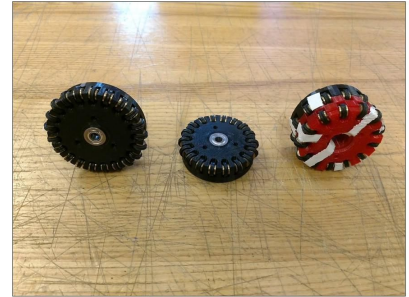
## Omni-Wheel Development



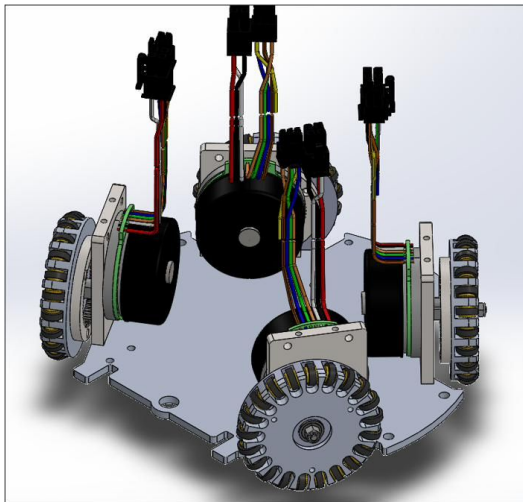
*Omni-wheel CAD — face view*



*Omni-wheel CAD — integrated gear*



*Physical wheel prototypes (iterations)*



*Full drivetrain assembly (CAD)*

## Contributions

- Redesigned omni-wheel geometry to improve agility and reduce mechanical play.
- Organised sub-projects targeting overall chassis rigidity.
- Facilitated design reviews for feasibility analysis and peer feedback.
- Produced engineering drawings for manufacturing.



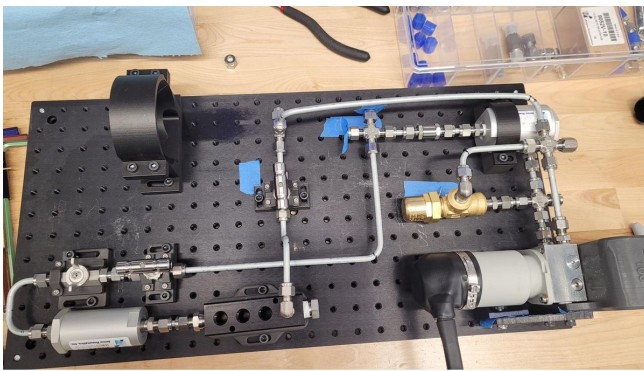
# Automated Test Infrastructure — Humanoid Robotic Hands

Sanctuary AI | Jan 2023 – Sept 2023 | Hardware Co-op: R&D

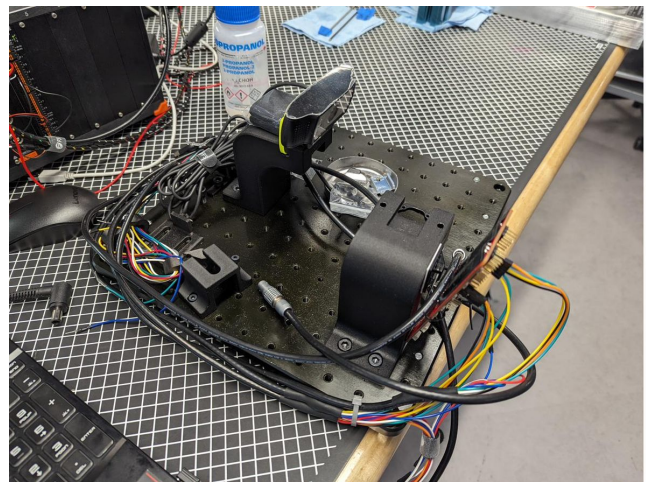
As a hardware R&D co-op, I developed automated test infrastructure for evaluating early-stage humanoid robotic hand prototypes — combining mechanical fixturing, electrical integration, data acquisition, and statistical analysis to accelerate the evaluation cycle. Descriptions are kept at a high level.

## Scope of Work

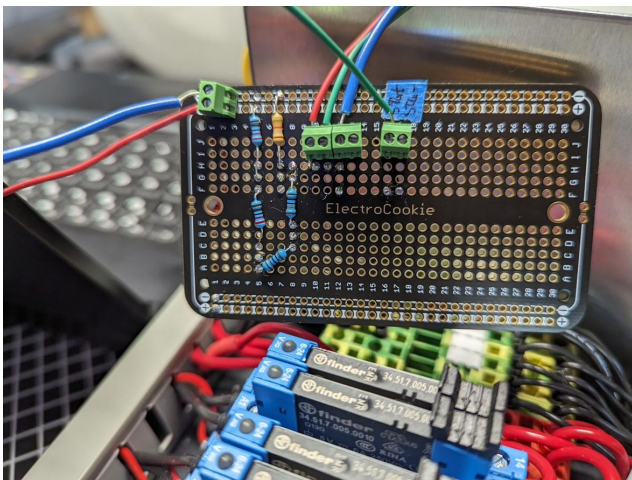
- Designed and built automated test rigs that raised prototype evaluation throughput by roughly 300%.
- Integrated force sensors, motors, and DAQ systems; wrote Python and C++ for data acquisition and PWM actuator control.
- Ran statistical analysis across 20+ prototype variants to identify failure modes; assembled and debugged relay control panels, hydraulic test benches, and custom fixturing.



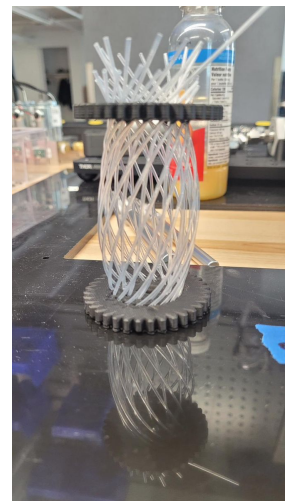
*Hydraulic/actuation test bench with instrumentation*



*Sensor-instrumented test fixture on optical breadboard*



*Custom control board & relay wiring*



*3D-printed compliant test component*



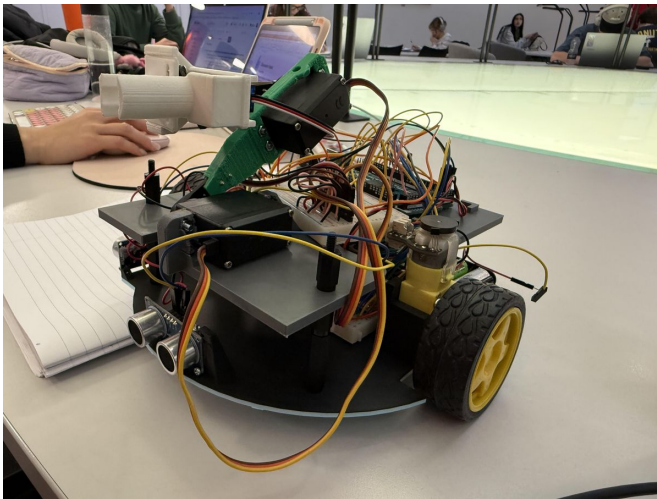
# Autonomous Mobile-Manipulator Robot

ELE435 Mechatronics — University of Sheffield | 2025/26

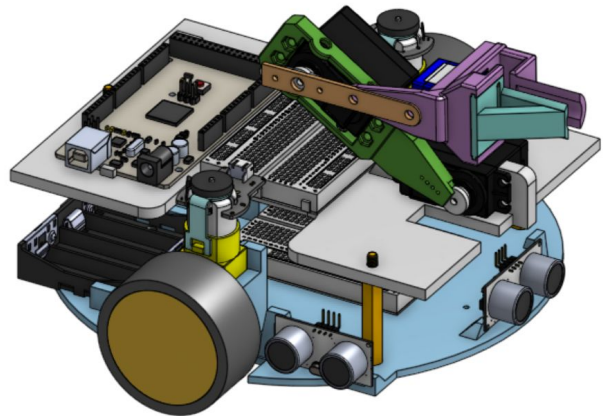
A fully autonomous mobile-manipulator robot designed and built from scratch for the ELE435 Mechatronics Robot Challenge. The robot navigates a structured arena — performing a 360° rotation, traversing two tunnels, drawing on a whiteboard with an onboard arm, and dropping a marker — all without human intervention. I led the mechanical design, 3D printing, component selection, electrical debugging, and the control state machine.

## System Highlights

- Mechanical design from scratch (Onshape) with fully 3D-printed custom structure.
- TB6612FNG-driven differential drive with quadrature-encoder angle control.
- Dual HC-SR04 ultrasonic sensors for reactive wall-following and tunnel traversal.
- 3-servo arm (2-DOF + end-effector) for whiteboard drawing and marker release.
- 8-state finite state machine implemented in Arduino C++.



Physical robot prototype — drivetrain, breadboard, and 3-servo arm



Full CAD assembly (Onshape) — arm, sensors, and electronics

**Demo video:** [youtu.be/LvyPOBPngcs](https://youtu.be/LvyPOBPngcs) — full autonomous run through the arena.

Full write-up, code, and CAD available at [github.com/SelloutCZN](https://github.com/SelloutCZN)